

Session 3: Story & Transform

From a fairness finding to a documented decision

QUEST · ICOTS-12 Pre-Conference Workshop

1:15 – 3:00

Brisbane, Australia

Where We Are

Time	Session	QUEST
9:00 – 10:30	Session 1: Question & Understand	Q, U
11:00 – 12:30	Session 2: Explain	E
1:15 – 3:00	Session 3: Story & Transform	S, T

We now have models we can explain. This session asks the harder questions.

Session 3 Roadmap

- | | | |
|-----|--------------------------------|-----------------------------------|
| 3.1 | Who does the model fail? | <i>icarm, fairmodels</i> |
| 3.2 | Telling the story honestly | <i>the Four C's</i> |
| 3.3 | Does this generalise? | <i>Mathematics vs. Portuguese</i> |
| 3.4 | The cut-off is a policy choice | <i>thresholds</i> |
| 3.5 | The actual deliverable | <i>audit trail</i> |
| 3.6 | Your turn | <i>capstone</i> |

`m_rf`, `m_cart`, and `explainer_rf` — fitted and explained in Session 2, on the same 295/100 student split.

We now put those explanations to work.

3.1 · Who Does the Model Fail?

PROBLEM

Session 2 asked whether we could trust the model's reasoning. It didn't ask whether that reasoning treats everyone the same way.

3.1 · A Group Fairness Audit

METHOD

```
fair_sex <- icarm_fairness(m_rf, test_bin, outcome = "pass",  
                           protected = "sex", positive = "Pass")
```

Split the test set by sex, and compute accuracy and catch-rate inside each group separately.

3.1 · The Headline Finding

RESULT

VERIFIED

Group	n	Accuracy	Actual Fails caught
Female	56	0.625	2 of 20 (10%)
Male	44	0.750	5 of 13 (38.5%)

3.1 · What This Means

INSIGHT

If this triggered a real support intervention, 9 of 10 female students who actually needed it would never be flagged.

This is not a claim about the SIZE of the gap being unusual — it's a claim about what the number actually means for real students, stated in plain language.

3.1 · A Second, Independent View

METHOD

```
fobject <- fairmodels::fairness_check(  
  explainer_rf, explainer_gbm, protected = test_bin$sex, privileged = "M")
```

icarm audits one model deeply. fairmodels compares several models on identical metrics, at once.

3.1 · Two Models, Same Metrics

RESULT

VERIFIED



3.1 · Accuracy and Fairness Don't Always Agree

INSIGHT

Model	Accuracy	fairmodels total loss
Random Forest	0.68	0.925 (better)
GBM	0.69	1.116

The slightly more accurate model is the slightly less equitable one. Checking both, on the same held-out data, is the only way to know which is true for a given pair of models.

3.2 · Telling the Story Honestly

PROBLEM

Every result in this workshop so far is a number. A number, told badly, invites the listener to invent their own story — usually the more dramatic one.

3.2 · The Four C's

FRAMEWORK

- C** Claim The exact number, precisely stated — not a rounded impression
- C** Context What it means for a real decision
- C** Case One concrete, specific example — a real row, not an aggregate
- C** Caveat A genuine limitation, every time

3.2 · Applied to Section 3.1

FRAMEWORK

Claim	The model catches 10% of actually-failing female students, vs. 38.5% of male students.
Context	9 of 10 female students who need support would never be flagged.
Case	2 of 20 actually-failing female students caught, vs. 5 of 13 male.
Caveat	One 100-student test split — worth confirming on a fresh sample.

3.2 · Bad, Better, Best

Bad

Random Forest produced 68% accuracy.

Better

Students with repeated failures are at greatest risk of poor outcomes.

Best

Early academic support for students with previous failures may help more than focusing on attendance alone — failures outweighs every other variable, in every model family we tried.

YOUR TURN

Using the Section 3.1 fairness finding, write ONE headline each for:

(a) a teacher (b) a parent (c) a school administrator

Same evidence. Three framings. None should claim more than the evidence supports.

3.3 · Does This Generalise?

PROBLEM

Everything so far comes from one course, Mathematics. Does any of it hold for a different subject?

3.3 · The Same Pipeline, a Different File

METHOD

```
run_subject <- function(file, label) {  
  df <- read.csv2(file, stringsAsFactors = TRUE)  
  df$pass <- factor(ifelse(df$G3 >= 10, "Pass", "Fail"))  
  sp <- icarm_split(df, prop = 0.75, seed = 2025, stratify = "pass")  
  # ... identical code from here on  
}  
run_subject("data/student-mat.csv", "Mathematics")  
run_subject("data/student-por.csv", "Portuguese")
```

3.3 · Same Data Source, Two Courses

RESULT

VERIFIED

	Mathematics	Portuguese
Students	395	649
Pass rate	67.1%	84.6%
Tree's first question failures, in BOTH subjects		

3.3 · What Stays the Same

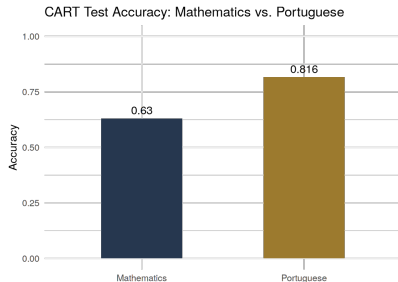
INSIGHT

The tree's first question, top 3 importance, and overall philosophy of the model transfer directly. That part generalises.

3.3 · What Changes

RESULT

VERIFIED



Higher accuracy for Portuguese isn't a better model — it's a higher pass rate. Specificity gets WORSE (0.24 to 0.12): the model catches even fewer students who go on to fail.

3.3 · The Fairness Gap Changes Shape

RESULT

VERIFIED

	Female catch-rate	Male catch-rate
Mathematics	10%	38.5%
Portuguese	0%	0%

INSIGHT

In Mathematics, the model is unfair BETWEEN groups. In Portuguese, it is equally useless FOR both groups — it catches zero actual failures, for anyone.

Same code, same method, a completely different failure mode. Generalising a finding means testing it, not assuming it.

3.4 · The Cut-Off Is a Policy Choice

PROBLEM

Every model so far used the default rule: predict Pass if predicted probability > 0.50 . Who decided that number was right?

3.4 · Sweeping the Threshold

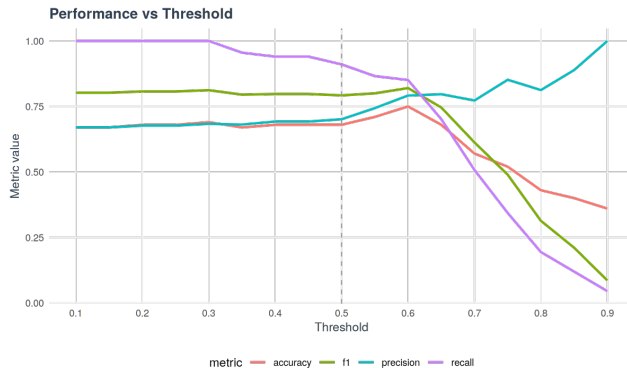
METHOD

```
probs <- predict(m_rf, test_bin, type = "prob")[, "Pass"]  
thr <- icarm_thresholds(test_bin$pass, probs, positive = "Pass")  
icarm_plot_thresholds(thr, metrics = c("accuracy", "recall", "precision", "f1"))
```

3.4 · Four Metrics, Every Cut-Off

RESULT

VERIFIED



icarm

INSIGHT

Accuracy peaks near 0.60, not 0.50. Recall and precision trade off in opposite directions on either side of it.

Missing a struggling student and wrongly flagging a fine one cost different things, to different people. `icarm` can show you the trade-off. It cannot make the choice for you.

3.5 · The Actual Deliverable

PROBLEM

Every finding in this workshop lives in a slide deck. A slide deck is not evidence anyone else can check.

3.5 · A Reproducible Record

METHOD

```
icarm_audit(m_rf, metrics = metrics_rf, fairness = fair_sex,  
            analyst = "Your Name", path = "outputs/audit_trail.json")  
icarm_scorecard(m_rf, test_bin, outcome = "pass", protected = "sex",  
                positive = "Pass", project = "ICOTS-12 Workshop",  
                path = "outputs/scorecard.json")
```

3.5 · What Gets Written

RESULT

VERIFIED

```
"accuracy": 0.680, "balanced_acc": 0.561,  
"max_acc_gap": 0.125, "max_eo_gap": 0.285,  
"disparate_impact_pass": true
```

INSIGHT

**A timestamped file.
Not a claim.**

Anyone — a colleague, a school board, a future you — can re-open this file and check every number in this session against it. That is the actual difference between an analysis and an accountable decision.

YOUR TURN

Working in your pair, pick ONE attribute — different from sex — and repeat the fairness audit yourself.

3.6 · Choose Your Attribute

Attribute	What it represents
Pstatus	Parents living together or apart
internet	Internet access at home
schoolsup	Extra educational support from the school
guardian	Student's guardian (mother, father, other)

No two pairs in the room should choose the same one — check with your neighbours.

3.6 · Run the Audit

```
ATTR <- "internet"    # your assigned attribute
team_fair <- icarm_fairness(m_rf, test_bin, outcome = "pass",
                           protected = ATTR, positive = "Pass")
icarm_plot_fairness(team_fair, metric = "acc")
```

3.6 · Prepare Your Three Minutes

Claim	Your exact fairness numbers
Context	What it means for a real decision
Case	One specific student from your test set
Caveat	A genuine limitation
Recommendation	Deploy as-is / with safeguards / not at all

One Session, Six Real Findings

- 3.1 A real, sizeable fairness gap by sex — and a second model that's less accurate but more equitable
- 3.2 The Four C's, applied to that finding
- 3.3 The finding partly generalises, partly doesn't — across two subjects
- 3.4 The decision threshold is a policy choice, not a default
- 3.5 A reproducible file, not a slide, is the real deliverable
- 3.6 Your own case, audited the same way

Claim. Context.
Case. Caveat.

Every result you present, for the rest of your career, deserves all four.

Thank you.

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